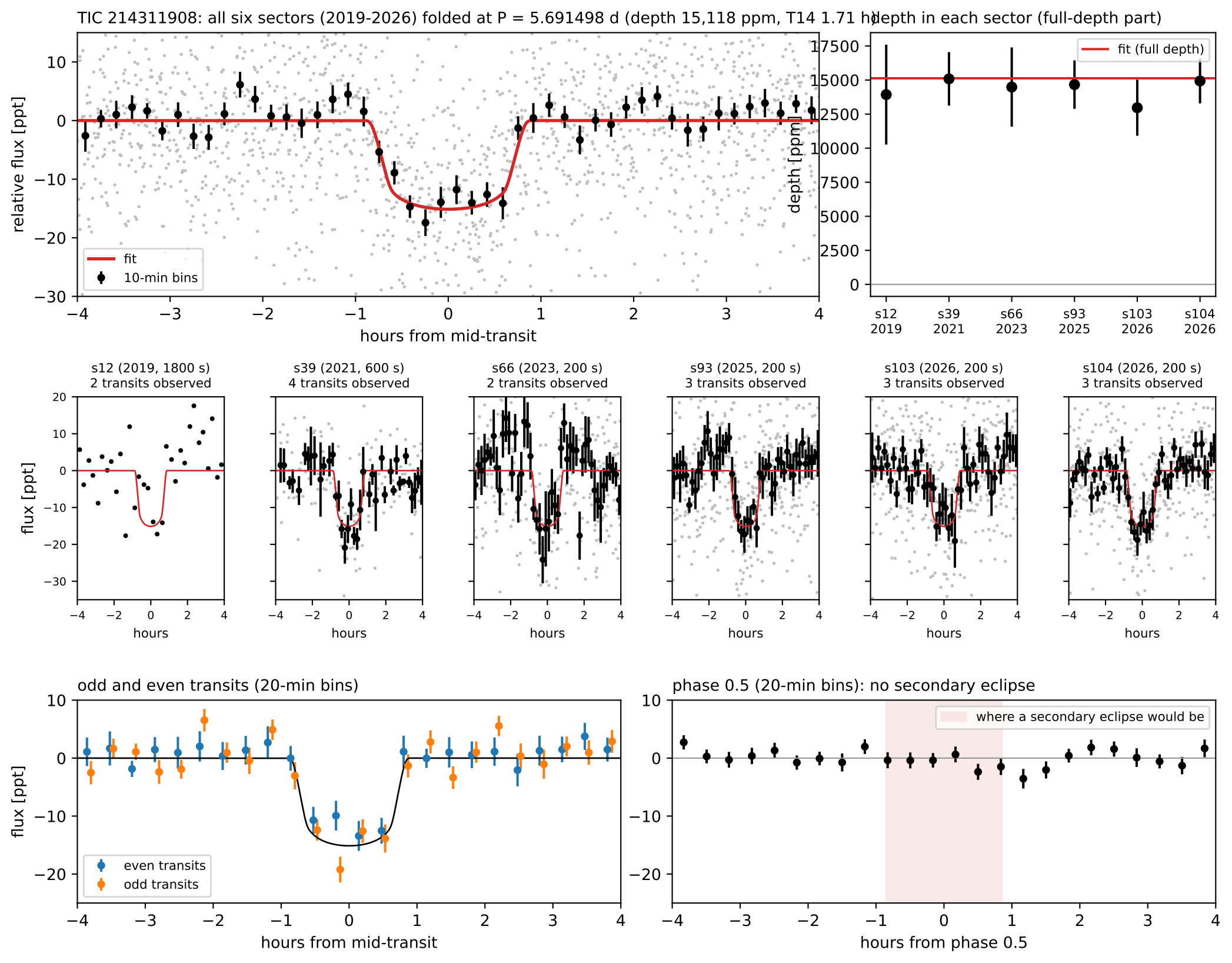
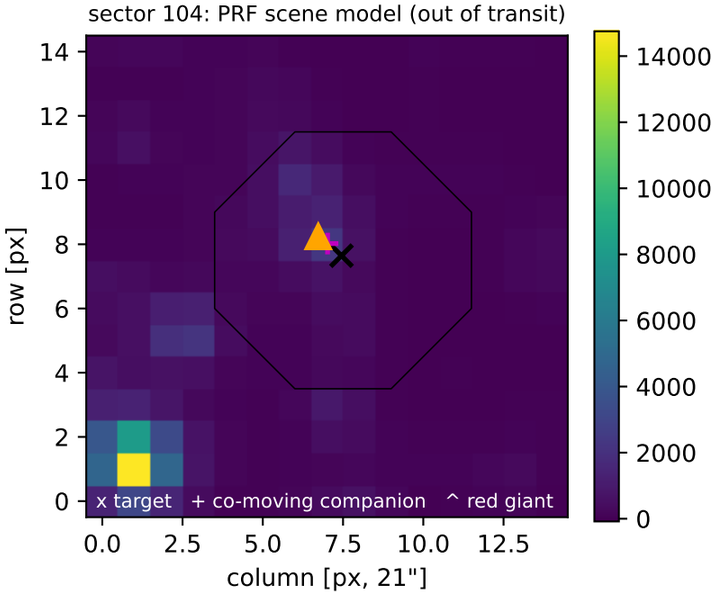
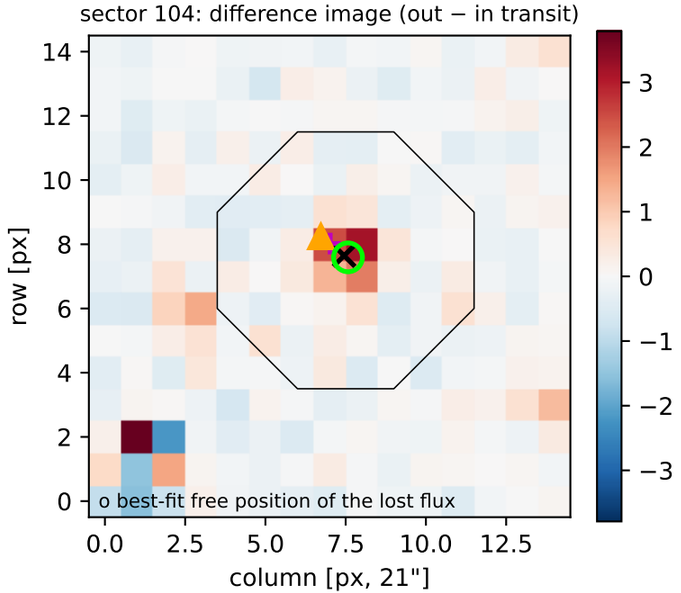
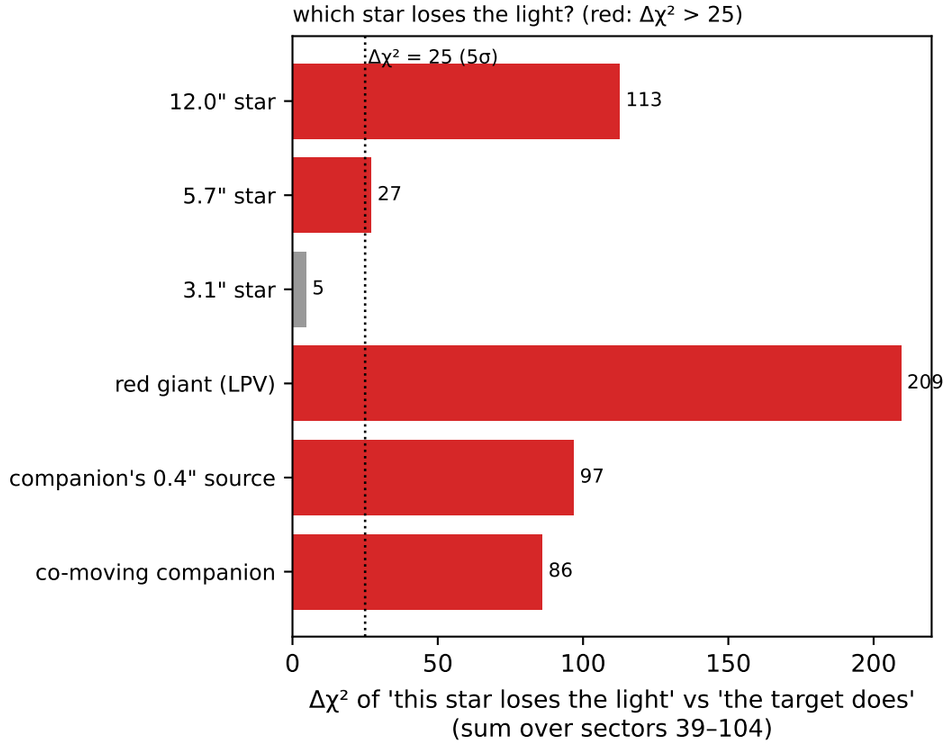
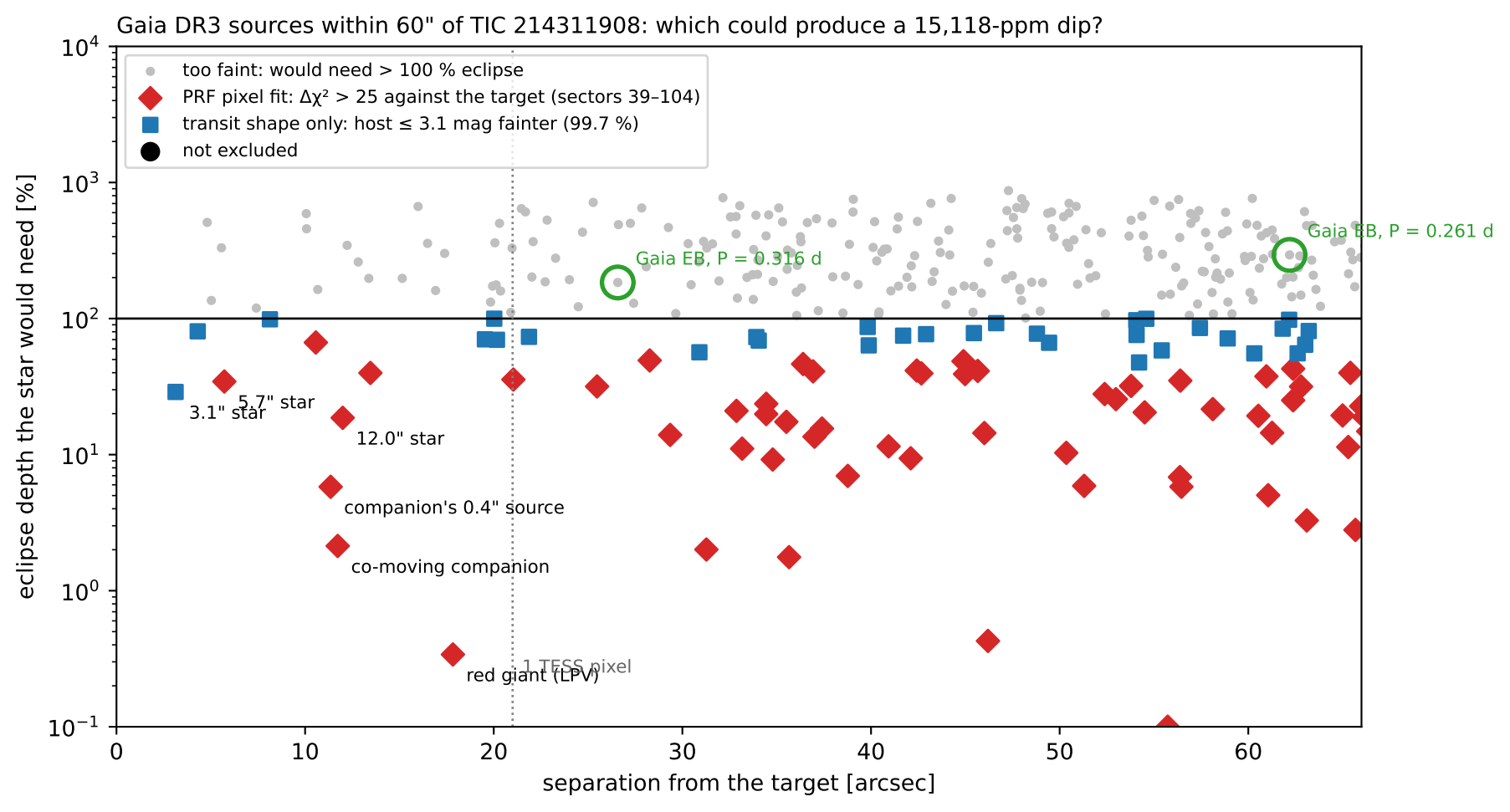

TIC 214311908: all six sectors (2019-2026) folded at P = 5.691498 d (depth 15,118 ppm, T14 1.71 h)
relative flux [ppt]
hours from mid-transit
fit
10-min bins
depth in each sector (full-depth part)
depth [ppm]
fit (full depth)
s12
2019
s39
2021
s66
2023
s93
2025
s103
2026
s104
2026
s12 (2019, 1800 s)
2 transits observed
s39 (2021, 600 s)
4 transits observed
s66 (2023, 200 s)
2 transits observed
s93 (2025, 200 s)
3 transits observed
s103 (2026, 200 s)
3 transits observed
s104 (2026, 200 s)
3 transits observed
flux [ppt]
hours
odd and even transits (20-min bins)
even transits
odd transits
hours from mid-transit
phase 0.5 (20-min bins): no secondary eclipse
where a secondary eclipse would be
hours from phase 0.5


sector 104: PRF scene model (out of transit)
x target + co-moving companion ^ red giant
row [px]
column [px, 21"]


sector 104: difference image (out − in transit)
o best-fit free position of the lost flux
row [px]
column [px, 21"]


which star loses the light? (red: Δχ² > 25)
Δχ² = 25 (5σ)
12.0" star
113
5.7" star
27
3.1" star
5
red giant (LPV)
209
companion's 0.4" source
97
co-moving companion
86
Δχ² of 'this star loses the light' vs 'the target does'
(sum over sectors 39–104)


Gaia DR3 sources within 60" of TIC 214311908: which could produce a 15,118-ppm dip?
too faint: would need > 100 % eclipse
PRF pixel fit: Δχ² > 25 against the target (sectors 39–104)
transit shape only: host ≤ 3.1 mag fainter (99.7 %)
not excluded
Gaia EB, P = 0.316 d
Gaia EB, P = 0.261 d
3.1" star
5.7" star
12.0" star
companion's 0.4" source
co-moving companion
red giant (LPV)
1 TESS pixel
eclipse depth the star would need [%]
separation from the target [arcsec]
10^4
10^3
10^2
10^1
10^0
10^{-1}
0
10
20
30
40
50
60